Analysis & Prediction of Commodity Prices in Zambia using Machine Learning

The CRISP DM Methodology

Business Understanding: The goal of the analysis is to understand and predict commodity prices over time.

Data Understanding: Dataset contains attributes such as location, commodity category/name, price, month & year

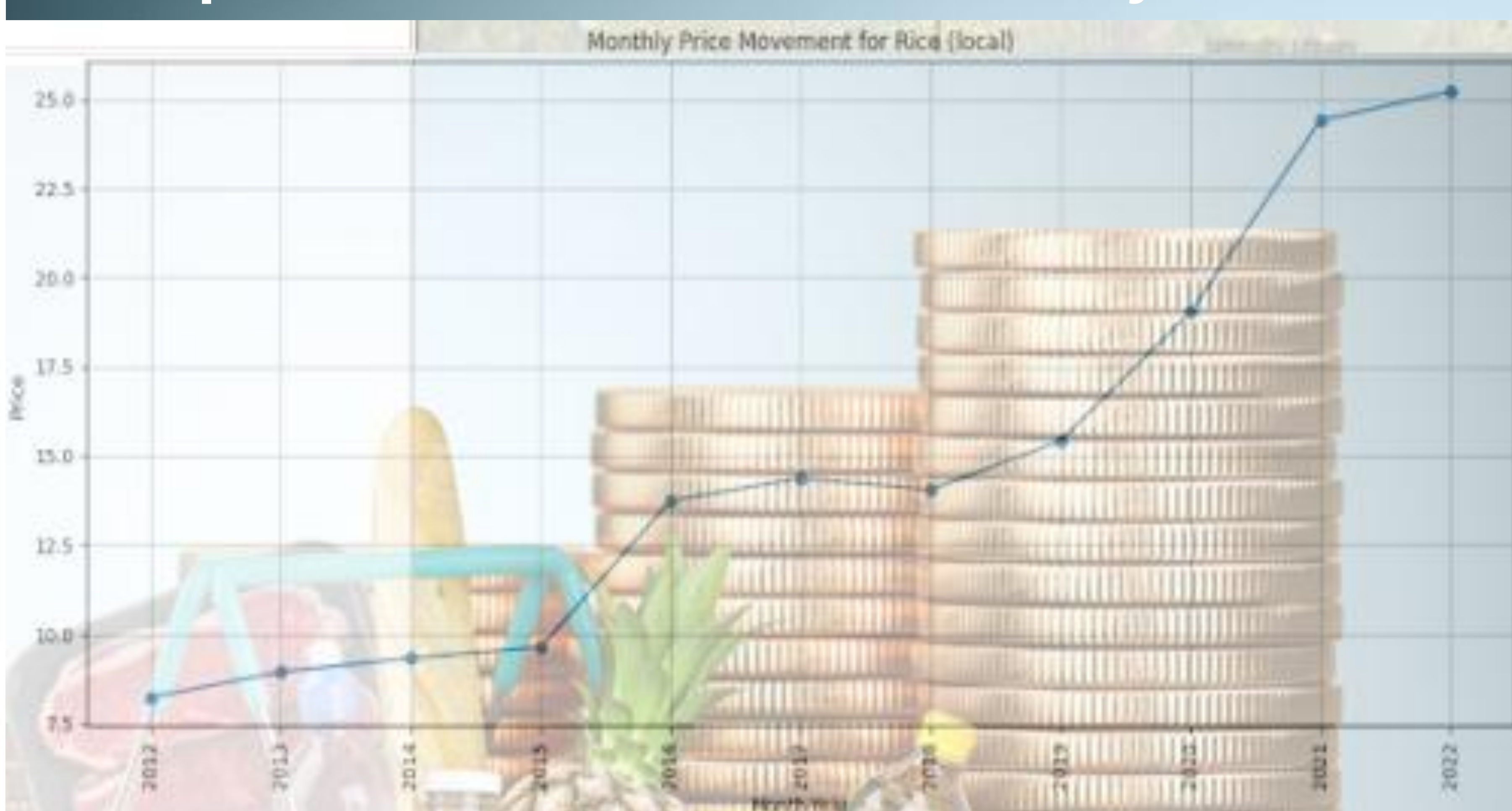
Data Preparation: Several steps are taken to pre-process the data for modeling including feature engineering.

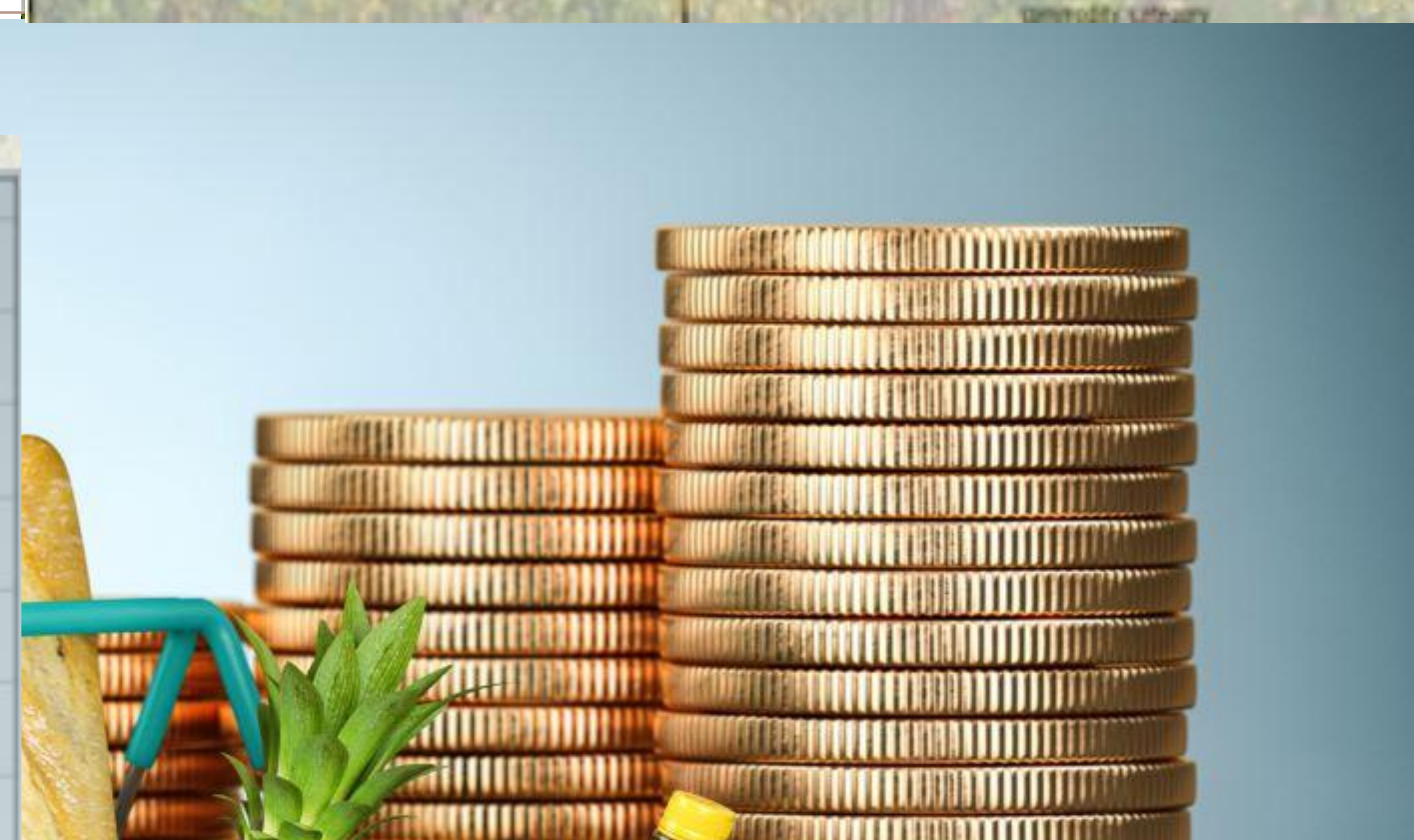
Modeling: A Random Forest Regressor is used to build the predictive model by Splitting the data, Training the model & making predictions of the commodity prices

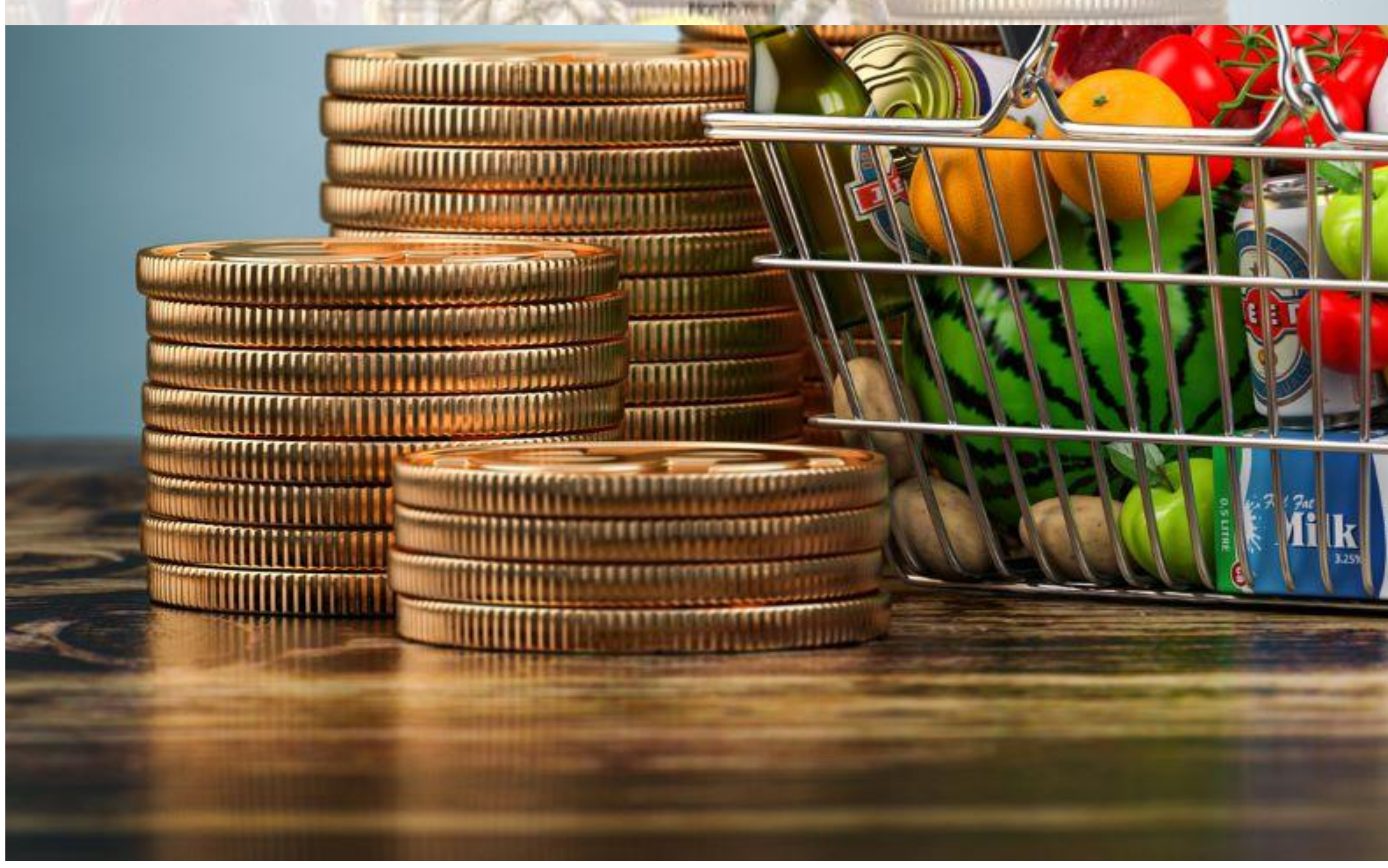
Commodity Categories Include:

- Cereals & Tubers
- Pulses & nuts
- Non-food
- Meat, Fish & Eggs
- Milk & Dairy
- Miscellaneous Food
- Oil & fats
- Vegetables & Fruits

The price movement of Rice over a 10-year Period







The Average survey duration was 29 Days

Prediction Model Evaluation

Features Selected:

- Geo Location (Latitude; Longitude)
- Commodity Category
- Commodity Name
- Month & Year

Evaluation Results

- Mean Squared Error: 9.54
- R-squared Score (Accuracy): **94%**
- Root Mean Squared Error (RMSE): 3.09
- Mean Absolute % Error (MAPE): 11.79